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Bayesian analysis. Final Project

Learning Route Choice via Thompson Sampling

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Abstract

Repeated decision-making under uncertainty arises in many real-world situations. Consider the problem of repeatedly travelling between two locations using one of several available routes. The central challenge is the exploration-exploitation trade-off: whether to choose the route that currently appears to be the best or to explore alternative routes.

This can be formulated as a Multi-Armed Bandit problem, where the travel time of each route is modeled as a random variable drawn from an unknown but fixed probability distribution.

Thompson Sampling is a learning algorithm that addresses this challenge by taking into account both current estimates of route performance and the uncertainty associated with those estimates.

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1 Introduction

This section introduces the problem addressed in this project and its motivation. It also presents the project objectives, the experimental setup used in the study, and the theoretical background needed to understand the proposed methodology.

The source code developed for this project is publicly available at https://github.com/MiguelDonado/ThompsonSampling_RouteChoice.

1.1 Description of the problem and justification

Many real-world problems involves making the same decision repeatedly under uncertainty. As experience is accumulated, the outcomes of past decisions can be used to improve future choices. Examples include selecting a restaurant, choosing a training plan, or deciding which route to take when traveling between two locations. The present project focuses on this last example, namely the route-choice problem.

Suppose that several alternative routes are available between two locations and that a driver must choose one of them on a daily basis. The travel time associated with each route is assumed to be a random variable drawn from an unknown but fixed probability distribution. In other words, although travel times vary from day to day, the probabilistic mechanism generating them is assumed to remain unchanged over time. The objective is to minimize the cumulative travel time over the entire decision horizon. If the underlying distributions were known in advance, the problem would be trivial, as one could simply choose the route with the lowest expected travel time at every decision step.

To fit the problem into the Multi-Armed Bandit framework, the reward is defined as the negative travel time. Therefore, maximizing reward is equivalent to minimizing travel time. The main challenge on this problem is to balance exploration and exploitation. On the one hand, different routes must be explored in order to gather information about their performance. On the other hand, once sufficient knowledge has been acquired, the driver should increasingly exploit the route that appears to be the best. An effective strategy should avoid excessive exploration of clearly suboptimal routes.

The motivation for this project stems from ongoing research on route-choice learning using the SUMO traffic simulator. In that setting, all drivers repeatedly choose among alternative routes and learn from the travel times they experience. One of the main goals is to study whether such learning processes converge to traffic equilibria.

This project considers a simplified version of the route-choice problem studied in the ongoing research by focusing on a single agent. An *agent* is the entity responsible for making decisions, which in this case corresponds to a driver choosing among alternative routes. Route travel times are assumed to follow unknown but fixed probability distributions, and route choices are made using the Thompson Sampling algorithm. This simplification allows the learning process to be studied independently of the interactions between multiple drivers.

The travel-time distributions considered in this work are not fixed properties of the routes themselves. They depend on factors such as departure time and the way background traffic is generated. As a result, different traffic scenarios could lead to different travel-time distributions for the same route.

1.2 Objectives

The main objectives of this project are:

- (1) Evaluate the ability of Thompson Sampling to identify the route with the lowest expected travel time while maintaining an efficient balance between exploration and exploitation.
- (2) Analyse the evolution of the posterior distributions and evaluate whether the algorithm can identify the optimal route without requiring highly accurate estimates of all routes.
- (3) Evaluate the performance of the algorithm using standard metrics from the Multi-Armed Bandit literature.

To evaluate the performance of Thompson Sampling, two experimental scenarios are considered throughout this project. Each scenario corresponds to a different traffic network, a different demand generation, and a different set of route alternatives available to the agent.

1.3 Experimental setup

Unlike traditional data-analysis workflows that rely on existing datasets, this project generates synthetic data using SUMO (Simulation of Urban MObility), an open-source traffic simulator. The following subsections describe the data-generation process.

1.3.1 Network configuration

As mentioned previously, two experimental scenarios are considered, each scenario based on a different network. The networks were constructed using the SUMO network editor (**NetEdit**). For consistency, they share several characteristics: all links consist of a single lane, no traffic lights or roundabouts are included, and a uniform speed limit of 50 km/h is imposed throughout the network. Furthermore, every node can potentially act as both an origin and a destination.

A summary of the main characteristics of the networks is provided in Table 1 while their graphical representations are shown in Figure 1.

TABLE 1. Summary of the networks used in the experiments.

Scenario	Network	Nodes	Edges	Turning movements	Avg. link length (m)
1	Toy	8	16	26	146.66
2	Sioux Falls	24	76	178	120.89

The first scenario uses a small toy network, whereas the second uses Sioux Falls, a well-known benchmark network in transportation research frequently used to evaluate traffic assignment and route-choice algorithms.

Route alternatives

For each scenario, a set of feasible routes is defined for the agent. These routes constitute the alternatives available to the agent.

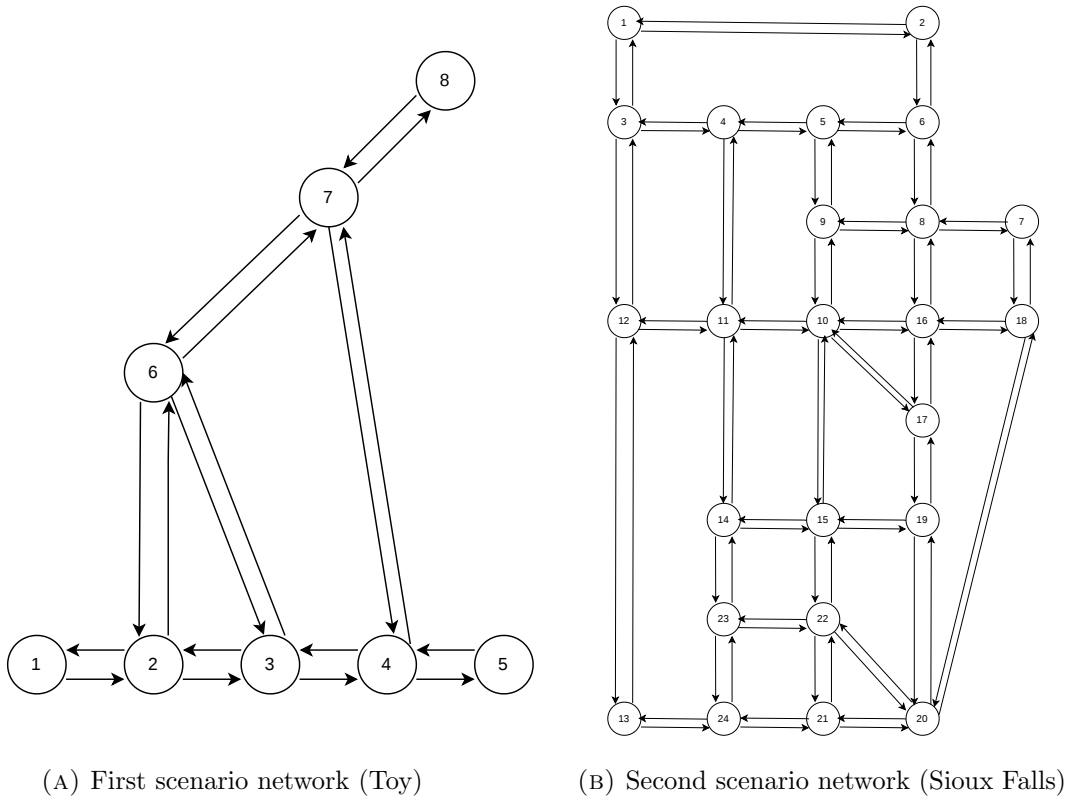


FIG. 1. Networks used in the project.

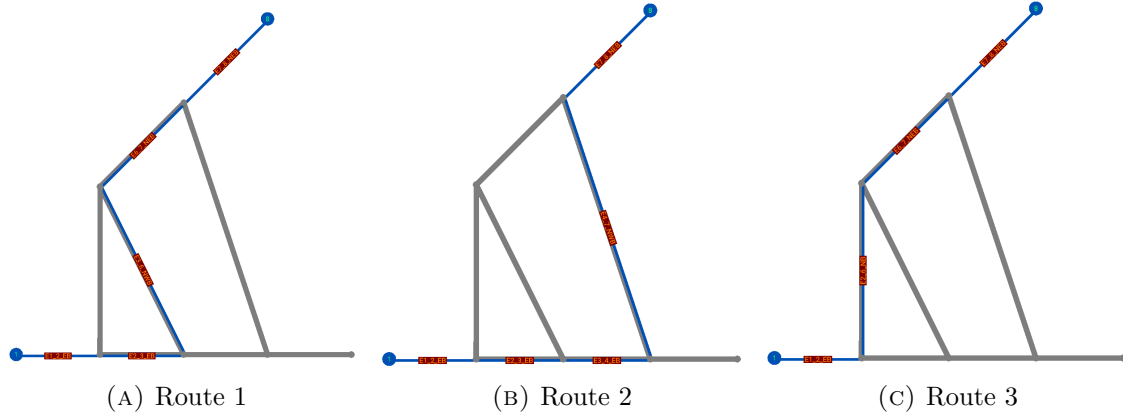


FIG. 2. Alternative routes considered in the network of the first scenario.

Figures 2 and 3 illustrate the route alternatives considered in the first and second networks, respectively.

Route characteristics

Tables 2 and 3 summarize the main characteristics of the available routes for the agent in each scenario.

The free-flow travel time corresponds to the travel time obtained when no background traffic is present. The mean and standard deviation of travel times were estimated using

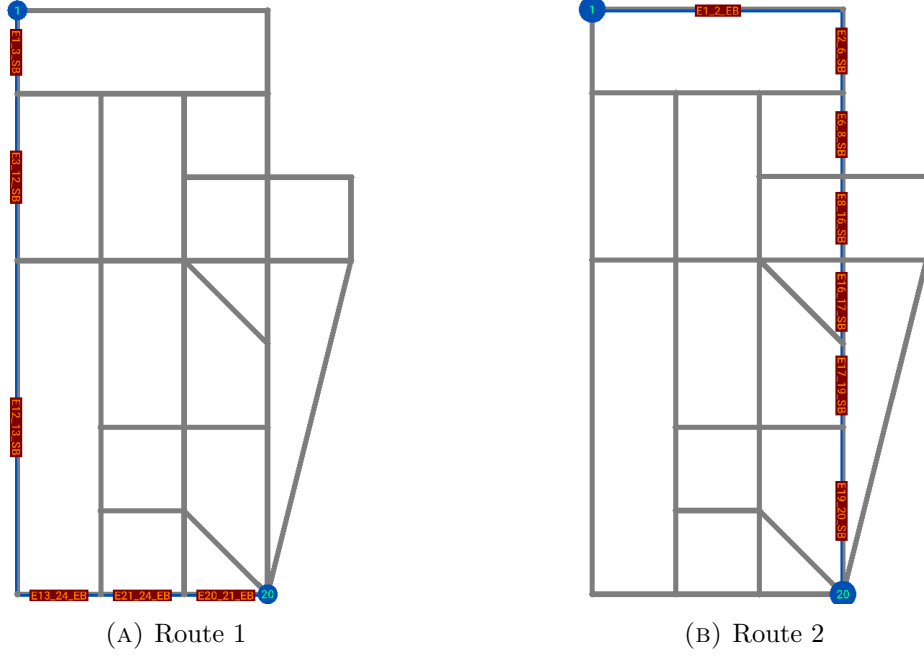


FIG. 3. Alternative routes considered in the network of the second scenario.

Monte Carlo simulation. For each route, 2000 simulations were performed with independently generated background traffic, and the resulting travel times were used to estimate the corresponding travel-time distribution.

As mentioned previously, these travel-time distributions depend on the experimental setting described in Section 1.3.2.

TABLE 2. Characteristics of the set of available routes for the agent in the first scenario.

Route	Length (m)	Free-flow TT (s)	Mean TT (s)	Std TT (s)
Route 1	529.24	38.10	58.72	13.09
Route 2	696.05	50.11	64.04	6.78
Route 3	634.47	45.67	79.97	23.56

TABLE 3. Characteristics of the set of available routes for the agent in the second scenario

Route	Length (m)	Free-flow TT (s)	Mean TT (s)	Std TT (s)	Gap (s)
Route 1	921.07	66.31	78.48	11.43	0.00
Route 2	885.75	63.77	80.13	11.81	1.65

1.3.2 Demand generation

Demand is generated using the SUMO traffic simulator. The default SUMO vehicle configuration is used throughout the experiments, allowing the focus to remain on route-choice behavior rather than on the effects of different vehicle characteristics.

Background Traffic Generation

In SUMO, traffic demand is defined by specifying the origin edge, destination edge, and departure time of each vehicle. To generate background traffic, the `randomTrips` tool is employed. This tool randomly assigns origins and destinations to vehicles, giving every edge the same probability of being selected, while departure times are generated uniformly throughout the simulation period.

Once all trips have been generated, the `duarouter` tool computes the shortest path for each vehicle. As a result, the background traffic is completely determined by SUMO’s demand generation and routing tools.

The background traffic is generated independently in each episode, where an episode corresponds to a single simulation run. As a result, traffic conditions vary across episodes, causing the travel times experienced by the agent to vary as well.

Table 4 summarizes the parameters used to generate the background traffic in each scenario.

TABLE 4. Background traffic generation parameters.

Parameter	Scenario 1	Scenario 2
Simulation duration (s)	540	540
Number of background vehicles	156	467

Agent

In addition to the background traffic, a single agent is introduced into the network. The origin, destination, departure time and set of available routes for this agent are fixed beforehand and remain unchanged throughout the experiments. In each episode, the agent selects a route using Thompson Sampling.

Table 5 summarizes the parameters defining the agent in each network.

TABLE 5. Agent parameters.

Parameter	Scenario 1	Scenario 2
Departure time (s)	360	360
Origin node	1	1
Destination node	8	20
Number of route alternatives	3	2

1.3.3 Simulation outputs

This project uses the SUMO `tripinfo` output, which records the trip duration of every vehicle. The agent’s trip duration is used as its travel time.

1.3.4 Experimental workflow

The overall experimental procedure follows a day-to-day learning framework. At the beginning of each episode, a new realization of the background traffic is generated. The agent

then selects a route according to the Thompson Sampling algorithm and experiences the corresponding travel time. This observation is used to update the posterior distribution associated with the selected route. The process is repeated over the entire simulation horizon.

1.3.5 Example of generated data

Table 6 presents a small sample of the observations generated during the experiments. Each observation corresponds to a single episode and records the route selected by the agent together with the travel time experienced on that episode.

TABLE 6. Example of generated observations.

Episode	Route	Travel Time (s)
1	2	73.4
2	1	61.8
3	1	59.7
4	3	91.5
5	1	62.1

Additional variables used in the analysis, such as regret and posterior summaries, are computed from these observations and are therefore not included in the raw dataset.

1.4 Theoretical overview

1.4.1 Multi-Armed Bandit Problems

Multi-Armed Bandit problems arise in situations where an agent must make repeated decisions over time under uncertainty, while learning from the outcome of previous decision. To introduce the concept, it is useful to consider the example that gave the problem its name.

A *one-armed bandit* is an old term used to refer to a slot machine. The term *armed* originates from the lever that had to be pulled to play the machine, while *bandit* reflects the perception that slot machines take players money.

A Multi-Armed Bandit problem can therefore be viewed as a gambler entering a casino and facing several slot machines. Each time a slot machine is played, it produces a random payout drawn from an underlying probability distribution. These distributions are assumed to be fixed but unknown to the gambler. If the distributions were known in advance, the optimal strategy would be trivial: always play the machine with the highest expected reward. In practice, however, the gambler must learn about the payout distributions through repeated interactions.

This creates a fundamental dilemma. On one hand, the gambler may choose to *exploit* the information gathered so far by repeatedly selecting the machine that has produced the highest observed rewards. On the other hand, the gambler may choose to *explore* other machines that have not been played very often, since they could potentially be better than the machine that currently appears to be the best. Balancing these two objectives gives rise to the well-known *exploration-exploitation trade-off*.

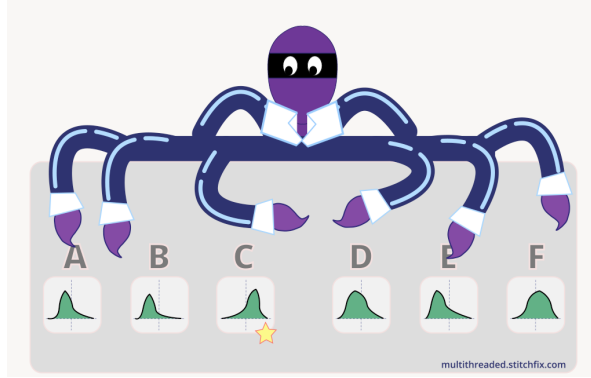


FIG. 4. Multi-Armed Bandit Problem. Source: Jana Beck, <https://multithreaded.stitchfix.com/blog/2020/08/05/bandits/>.

Consider a finite set of actions (or arms) $\mathcal{A} = 1, \dots, K$. At each decision step t , the agent selects an action $a_t \in \mathcal{A}$ and receives a stochastic reward R_t .

The expected reward of action a is

$$\mu_a = \mathbb{E}[R_t \mid a_t = a],$$

and the optimal action is

$$a^* = \arg \max_{a \in \mathcal{A}} \mu_a.$$

The objective of the agent is to maximize the cumulative reward over a horizon of T decision steps. Performance is commonly evaluated through regret. The instantaneous regret at time t is

$$r_t = \mu_{a^*} - \mu_{a_t},$$

while the cumulative regret after T steps is

$$R(T) = \sum_{t=1}^T r_t.$$

Regret quantifies the total loss incurred due to not always selecting the optimal action. Consequently, a desirable learning algorithm is one that achieves low cumulative regret. This project focuses on Thompson Sampling, a Bayesian approach to the Multi-Armed Bandit problem.

1.4.2 Thompson Sampling

To develop an intuition for Thompson Sampling, consider a Multi-Armed Bandit problem with K actions. Each action k produces a reward of one with probability θ_k and a reward of zero with probability $1 - \theta_k$. The reward associated with action k is therefore modeled as

$$R_t \mid a_t = k \sim \text{Bernoulli}(\theta_k),$$

where θ_k is unknown.

From a Bayesian perspective, uncertainty about θ_k is represented through a prior distribution. A natural choice is the Beta distribution,

$$\theta_k \sim \text{Beta}(\alpha_k, \beta_k).$$

Since the expected reward of a Bernoulli random variable equals its success probability,

$$\mu_k = \mathbb{E}[R_t \mid a_t = k] = \theta_k.$$

learning the expected reward of an action is equivalent to learning the value of the θ_k . After each decision, only the posterior distribution of the selected action is updated using the observed reward.

Suppose that after a number of observations have been collected, the posterior distributions shown in Figure 5 have been obtained.

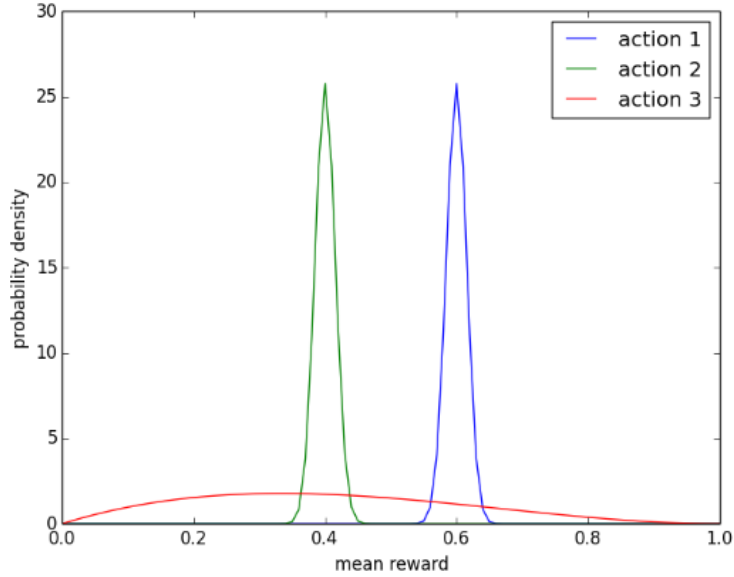


FIG. 5. Posterior distributions over expected rewards. Reproduced from [1]

A greedy strategy would select the action with the highest posterior mean, namely Action 1. However, it ignores the fact that Action 3 is much more uncertain and could still turn out to be the best alternative.

An ε -greedy addresses this issue by selecting the action with the highest posterior mean most of the time, while exploring a random action with probability ε . Although this ensures continued exploration, it treats all unexplored actions equally. As a result, it may spend resources on actions that already appear highly unlikely to be optimal. Ideally, exploration should focus on actions that still have a realistic change of being the best.

Thompson Sampling achieves this by using the posterior distributions themselves to guide exploration. At each decision step, it draws one sample from the posterior distribution of each action and selects the action with the largest sampled value.

In Figure 5, Action 1 is selected most of the time because it currently appears to be the best alternative. However, Action 3 is still selected occasionally because the available evidence is insufficient to rule out the possibility that it is the best action. In contrast, Action 2 is rarely selected because the available evidence strongly suggests that it is inferior to Action 1.

As a result, Thompson Sampling naturally balances exploration and exploitation. It mainly selects actions that are currently believed to be optimal while still allocating exploration to actions that could plausibly be optimal due to their uncertainty. Actions that are highly unlikely to be optimal receive little or no exploration. This behavior is often referred to as probability matching: actions are selected according to their posterior probability of being optimal.

It is worth emphasizing that, unlike the classical Bayesian workflow, where inference is performed on a fixed dataset, Thompson Sampling uses Bayesian inference to guide the data-collection process itself.

The Thompson Sampling algorithm can be summarized through the following steps:

- (1) Specify a probabilistic model for the rewards of each action and prior distributions for its unknown parameters.
- (2) Compute the posterior distributions using the observations collected so far.
- (3) Draw one sample from the posterior distribution associated with each action.
- (4) Use the sampled values to estimate the expected reward of each action.
- (5) Select the action with the highest sampled expected reward.
- (6) Observe the reward generated by the selected action.
- (7) Update the corresponding posterior distribution.
- (8) Repeat Steps 2–7 until the predetermined horizon T is reached.

2 Analysis

Since the objective is to minimize travel time, it is convenient to define the reward as the negative travel time

$$R = -TT$$

Under this definition, maximizing the expected reward is equivalent to minimizing the expected travel time.

2.1 Approximation of route travel-time distributions

Although Thompson Sampling does not require knowledge of the true travel-time distributions and instead learns from observed travel times, approximating these distributions is useful for several reasons. First, it allows the performance of the algorithm to be assessed using metrics such as regret. Second, it provides insight into the shape of the travel-time distributions, thereby helping to select an appropriate statistical model.

The travel-time distributions are approximated using Monte Carlo simulation. For each route, 2000 simulation runs are performed. Because a new background traffic realization is generated in every run, the observed travel times can be treated as samples from the corresponding travel-time distribution.

2.2 Model selection

Figures 6 and 7 show the empirical travel-time distributions obtained through Monte Carlo simulation. These distributions exhibit two important characteristics. First, travel times are strictly positive. Second, the distributions are positively skewed, with most observations concentrated around moderate travel times and a small number of observations corresponding to substantially larger delays caused by congestion.

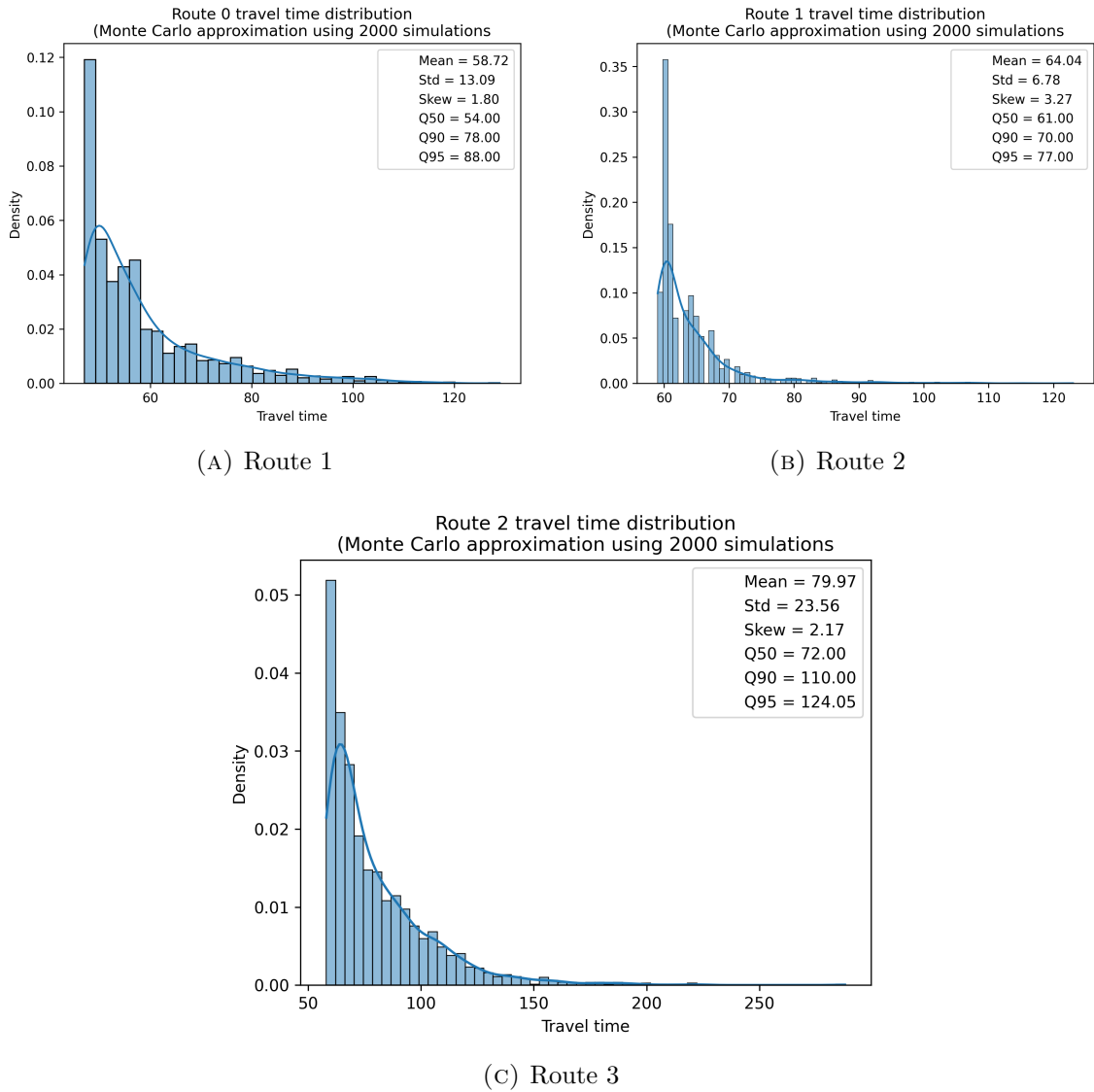


FIG. 6. Scenario 1. Empirical travel-time distributions obtained through Monte Carlo simulation. Histograms and kernel density estimates (KDEs).

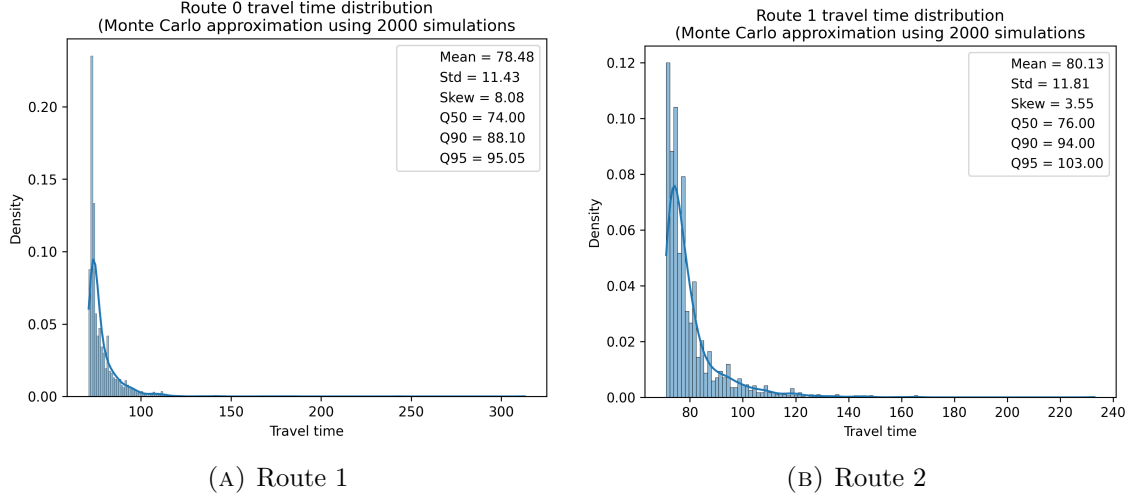


FIG. 7. Scenario 2. Empirical travel-time distributions obtained through Monte Carlo simulation. Histograms and kernel density estimates (KDEs).

These characteristics suggest that the Gamma distribution provides a suitable statistical model for route travel times.

2.3 Bayesian model and Thompson Sampling implementation

Travel times are modeled using a Gamma distribution,

$$TT \sim \text{Gamma}(\alpha, \beta),$$

where α denotes the shape and rate parameters, respectively. Under this parameterization,

$$\mathbb{E}[TT] = \frac{\alpha}{\beta}, \quad \text{Var}(TT) = \frac{\alpha}{\beta^2}.$$

In this work, the shape parameter α is assumed known, while the rate parameter β is treated as unknown. A Gamma prior is assigned to β ,

$$\beta \sim \text{Gamma}(a_0, b_0),$$

yielding a conjugate Bayesian model. Consequently, if

$$\beta \mid (TT_1, \dots, TT_{t-1}) \sim \text{Gamma}(a_{t-1}, b_{t-1}),$$

and a new travel-time observation is collected, the posterior hyperparameters are updated as

$$(1) \quad a_t = a_{t-1} + \alpha, \quad b_t = b_{t-1} + TT_t.$$

Therefore,

$$\beta \mid (TT_1, \dots, TT_t) \sim \text{Gamma}(a_t, b_t).$$

The conjugate model allows posterior distributions to be updated analytically after each new observation, making the model particularly suitable for the sequential learning setting considered in this work.

The value of α is estimated in advance using the Monte Carlo samples generated in the previous section. Although both Maximul Likelihood Estimation (MLE) and the Method of Moments are suitable alternatives, the latter is adopted due to its simplicity. Under the shape–rate parameterization,

$$\hat{\alpha}^{MoM} = \frac{\bar{x}^2}{s^2}$$

where $\bar{x}$ and s^2 denote the sample mean and sample variance of the Monte Carlo samples, respectively.

Table 7 reports the resulting estimates for each route.

Scenario	Route	$\hat{\alpha}_{\text{MoM}}$
First	1	20.13
First	2	89.21
First	3	11.51
Second	1	47.15
Second	2	46.05

TABLE 7. Method-of-moments estimates of the shape parameter α of the Gamma distribution for each route and scenario.

The prior hyperparameters were set to $a_0 = b_0 = 0.1$, resulting in a weakly informative prior.

At each episode, Thompson Sampling draws one sample from the posterior distribution of β for each route. These sampled values are then used to compute estimates of the expected travel times. The route with the lowest estimated expected travel time is selected. After observing the resulting time, the posterior distribution of the selected route is updated according to Equation 1. This process is repeated until the predetermined horizon T is reached.

The Bayesian models used for the different routes are summarized below.

Scenario 1

$$M_{\text{route1}} = \{p(TT \mid \theta) = \text{Gamma}(20.13, \beta_1), \beta_1 > 0\}, \quad \pi(\beta_1) = \text{Gamma}(0.1, 0.1)$$

$$M_{\text{route2}} = \{p(TT \mid \theta) = \text{Gamma}(89.21, \beta_2), \beta_2 > 0\}, \quad \pi(\beta_2) = \text{Gamma}(0.1, 0.1)$$

$$M_{\text{route3}} = \{p(TT \mid \theta) = \text{Gamma}(11.51, \beta_3), \beta_3 > 0\}, \quad \pi(\beta_3) = \text{Gamma}(0.1, 0.1)$$

Scenario 2

$$M_{route1} = \{p(TT \mid \theta) = \text{Gamma}(47.15, \beta_1), \beta_1 > 0\}, \quad \pi(\beta_1) = \text{Gamma}(0.1, 0.1)$$

$$M_{route2} = \{p(TT \mid \theta) = \text{Gamma}(46.05, \beta_2), \beta_2 > 0\}, \quad \pi(\beta_2) = \text{Gamma}(0.1, 0.1)$$

2.4 Learning performance of Thompson Sampling

This section examines the learning behavior of the Thompson Sampling algorithm in both experimental scenarios. The analysis focuses on the evolution of the posterior distributions, convergence towards the true mean travel times, and route-selection frequencies.

2.4.1 Posterior evolution

Before discussing the results, note that the posterior distributions shown for episode t correspond to the beliefs used to make the route choice at that episode, before observing the resulting travel time and updating the posterior.

The evolution of the posterior distributions of the mean travel times for Scenario 1 is shown in Figure 8. At episode 1, no travel-time observations have yet been collected, and therefore the posterior distributions coincide with the prior distributions. Since a weakly informative prior was used, there is initially considerable uncertainty about the expected travel times of all routes, resulting in broad posterior distributions.

By episode 2, the posterior distribution associated with Route 1 has already been updated, indicating that it was selected in the first episode. The distributions of the remaining routes remain unchanged due to the lack of observations. As a result, they are still equal to the weakly informative priors and therefore remain highly dispersed. Consequently, their densities are spread over a wide range of values and appear barely visible in the figure.

By episode 5 all routes have been selected at least once. Although Route 1 appears to be the most promising alternative, the posterior distributions of Routes 2 and 3 still overlap considerably with it, implying that they retain a non-negligible probability of being optimal and will continue to be explored.

By episode 99, the posterior distribution of Route 3 is located substantially to the right of the other distributions, indicating that it is unlikely to be the optimal route. At this point Route 1 is regarded as the most promising alternative, although there remains some uncertainty about its true mean travel time.

From episode 199 onwards, the posterior distribution of Route 1 becomes increasingly concentrated around its true mean travel time, providing strong evidence that it is the optimal route. As a result, Route 1 is selected most of the time, while exploration of the remaining routes becomes increasingly rare.

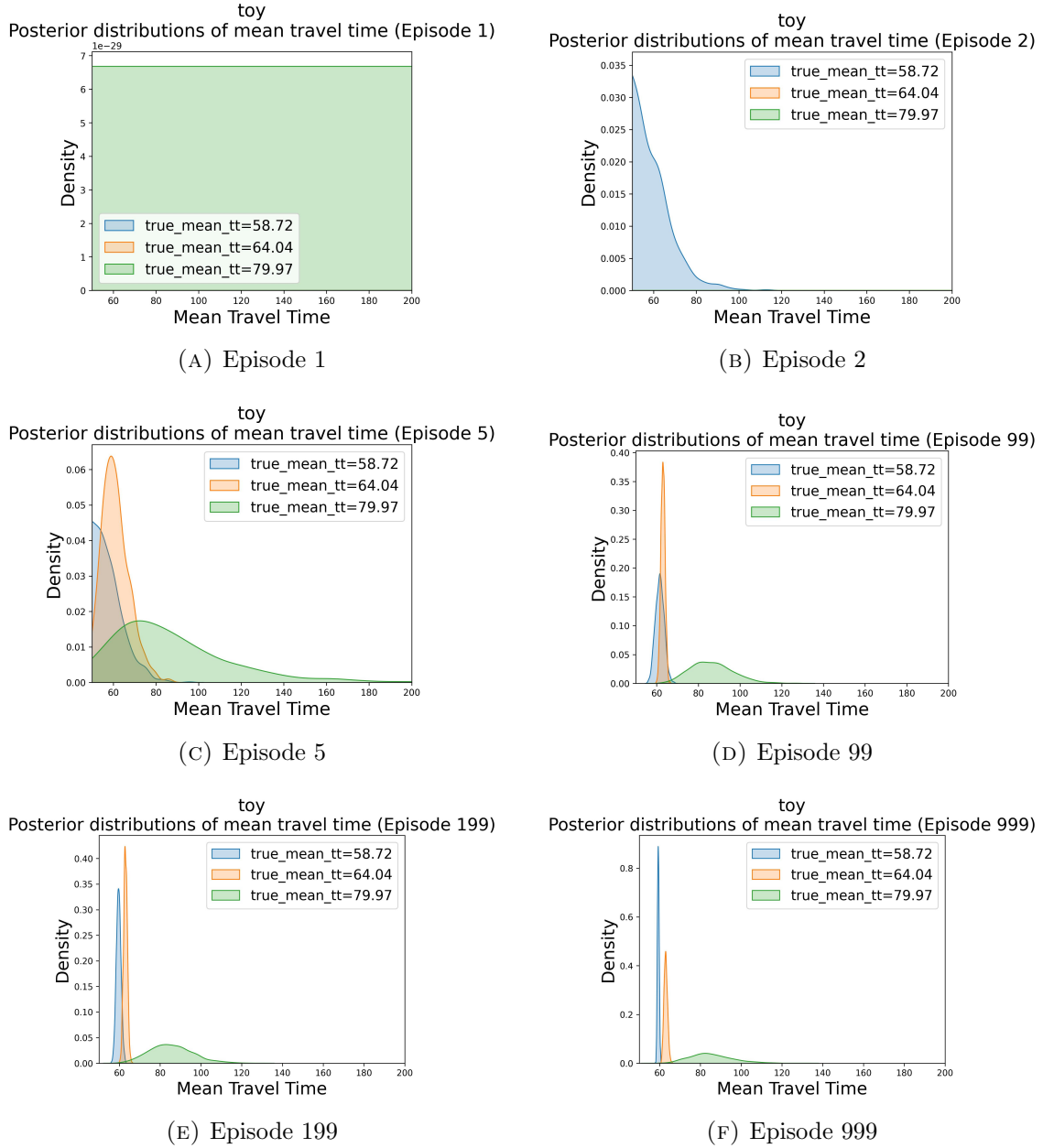


FIG. 8. Evolution of posterior distributions of mean travel times in the Toy network.

Figure 9 illustrates the evolution of the posterior distributions of the mean travel times for Scenario 2. As in the previous scenario, the posterior distributions initially coincide with the prior distributions. During the early stages of learning, Route 1 is selected repeatedly, and by episode 29 both routes have been selected.

At episodes 69 and 99, the posterior distributions of the two routes remain very similar. In fact, at episode 99 the algorithm temporarily favors Route 2, even though Route 1 has the lowest true mean travel time. By episode 299, Route 1 again appears to be the most promising alternative, although substantial overlap remains between the two distributions, showing that both routes remain plausible candidates for the optimal route.

Compared with Scenario 1, convergence is noticeably slower. This occurs because the true mean travel times are more similar and the corresponding true travel-time distributions exhibit greater overlap in Scenario 2. As a result, it is more difficult for the algorithm to determine which route is optimal.

From episode 299 onwards, the posterior distribution of Route 1 becomes progressively more concentrated around lower values, providing evidence that it is the optimal route. However, Route 2 still retains a non-negligible probability of being optimal, suggesting that additional observations would be needed to distinguish the two routes more clearly.

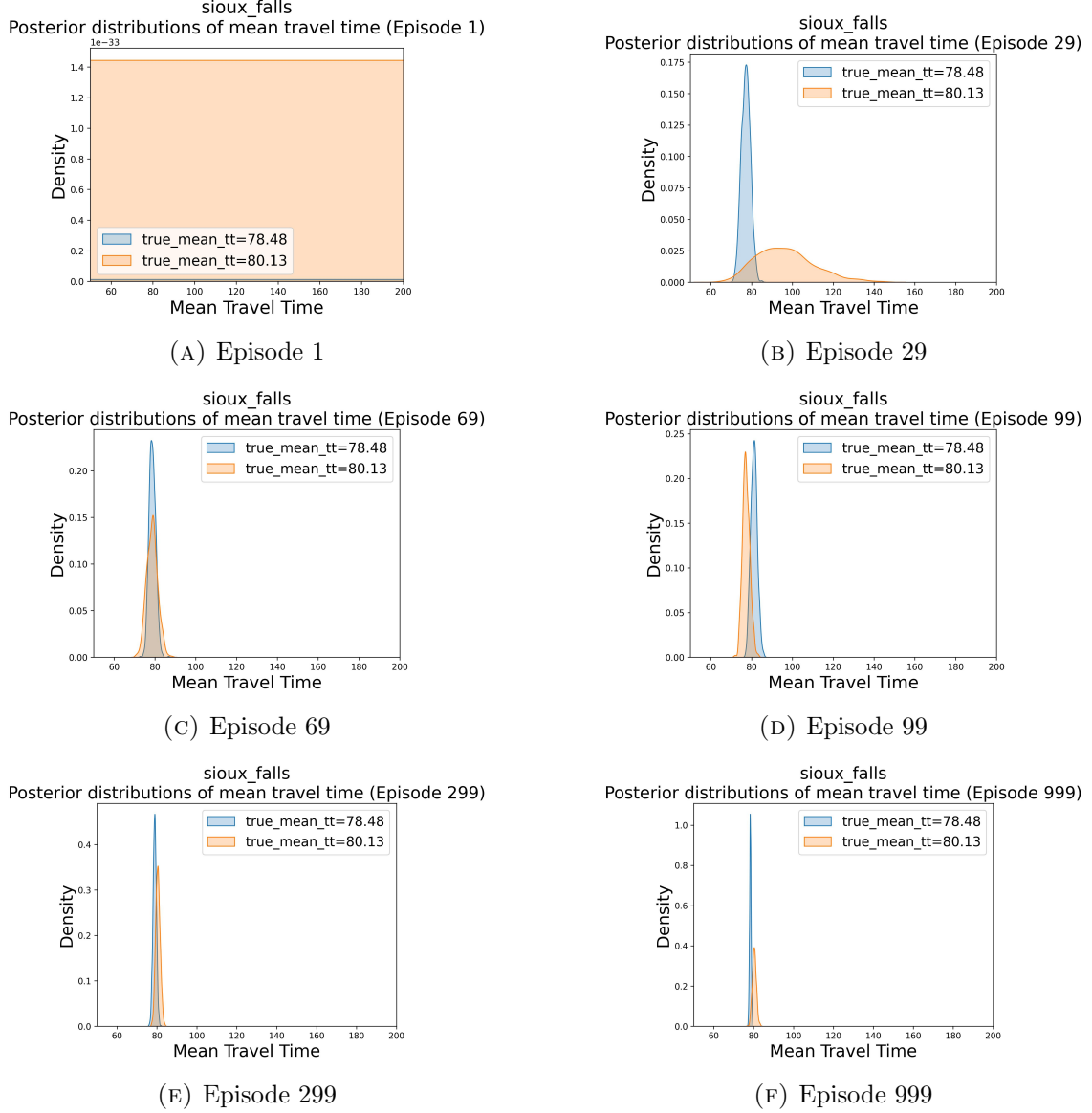


FIG. 9. Evolution of posterior distributions of mean travel times in the Sioux Falls network.

2.4.2 Posterior means vs true means

An important property of Thompson Sampling is that good decision-making does not require accurate estimation of the mean travel time of every route. Its objective is to

identify the route with the lowest expected travel time. Because each route is sampled according to its posterior probability of being optimal, Thompson Sampling collects more observations from routes that appear promising and fewer observations from routes that appear suboptimal. As a result, the mean travel time of the optimal route is typically estimated very accurately, whereas larger estimation errors may remain for suboptimal routes. In general, the worse a route is relative to the optimum, the fewer observations are needed to rule it out, and therefore the larger the estimation may be.

Figure 10 compares the posterior mean travel times obtained at the end of the learning process with the true mean travel times estimated through Monte Carlo simulation. Points lying on the diagonal correspond to routes whose posterior mean coincides with the true mean.

In Scenario 1, the posterior mean of Route 1, the optimal route, is almost identical to its true mean. Route 2, the second-best alternative, is also estimated with good accuracy. In contrast, Route 3 exhibits the largest discrepancy between its posterior mean and true mean. This behavior is consistent with the learning mechanism of Thompson Sampling: once sufficient evidence has been gathered to conclude that Route 3 is unlikely to be optimal, the algorithm allocates relatively few additional observations to it.

In Scenario 2, both routes are estimated accurately. Since their true mean travel times and variance are very similar, the algorithm requires relatively accurate estimates of both means in order to determine which one is optimal. Consequently, both routes continue to be sampled throughout the learning process, resulting in more accurate estimates.

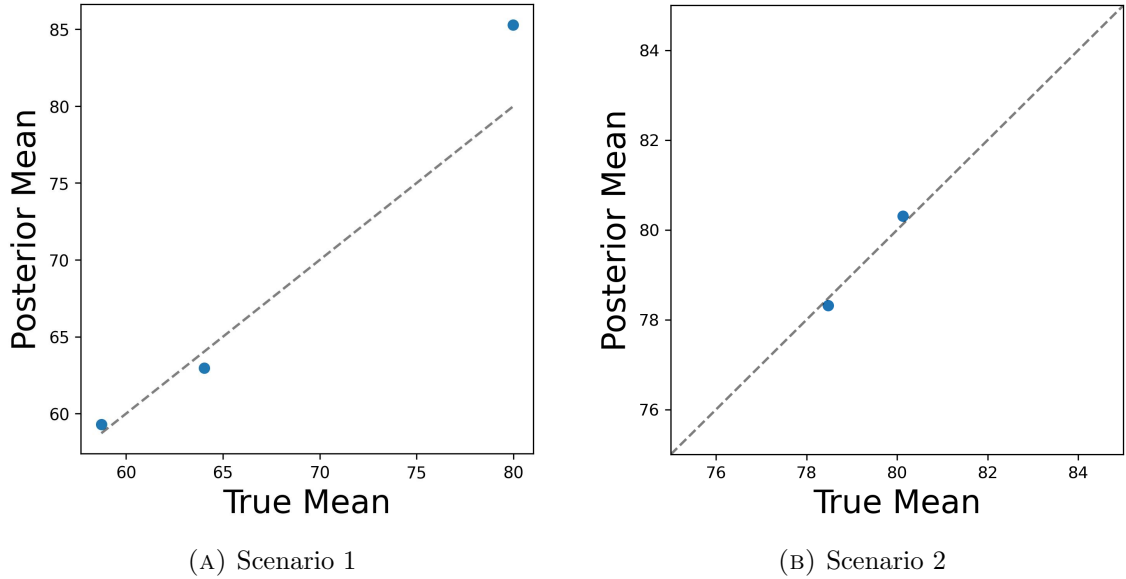


FIG. 10. Comparison of Posterior means vs True means

2.4.3 Route selection frequencies

This section analyzes how route-selection frequencies evolve over time. These frequencies illustrate how Thompson Sampling initially explores different alternatives and progressively exploits the routes that appear most promising.

Figures 11 present the cumulative route-selection frequencies for each route. The horizontal axis represents the episode number, while the vertical axis shows the proportion of episodes up to that point in which each route has been selected.

For Scenario 1, Route 3 is quickly identified as a poor alternative. After a brief exploration phase, it is selected only rarely, indicating that the algorithm has gathered enough evidence to conclude that it is unlikely to be the optimal route. During the first 100 episodes, the algorithm primarily explores Routes 1 and 2. From episode 100 onwards, Route 1 begins to dominate the selection frequencies, showing that the algorithm has identified it as the most promising alternative.

For Scenario 2, the routes have very similar travel-time distributions, and therefore the algorithm requires more observations to distinguish between them. Between episode 100-200, Route 2 is selected more frequently, suggesting that the algorithm temporarily considers it the most promising alternative. From approximately episode 200 onwards, as additional observations are collected, the evidence increasingly favors Route 1. Consequently, its cumulative selection frequency steadily increases, while the frequency of selecting Route 2 gradually declines.

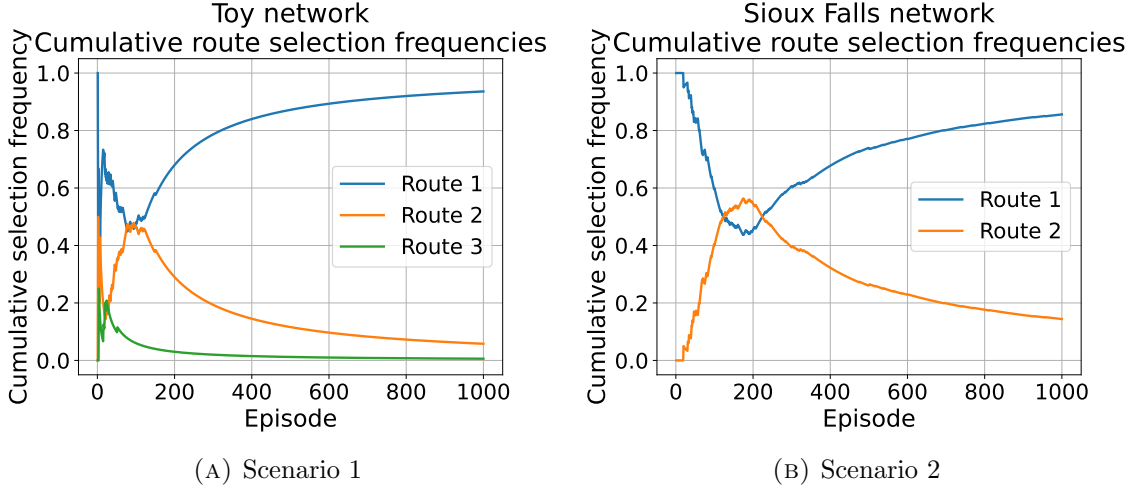


FIG. 11. Cumulative route-selection frequencies.

2.5 Regret analysis

This section analyze both the cumulative regret and the average regret per episode in order to assess the learning performance of Thompson Sampling in the two experimental scenarios.

Regret is one of the standard performance metrics in the Multi-Armed Bandit literature, as it measures the loss resulting from not always selecting the optimal action. Recall the definition of regret introduced in Section 1.4.1.

The cumulative regret measures the total loss accumulated over all episodes due to the exploration of suboptimal alternatives. An effective learning algorithm is expected to incur most of its regret during the initial stages of the experiment, when information about the available actions is limited and exploration is necessary. As learning progresses, the algorithm should select the optimal action more frequently, causing cumulative regret to grow at a much slower rate.

Consequently, a low cumulative regret indicates that the algorithm was able to gather strong evidence in favor of the optimal action relatively quickly and therefore spent less time exploring suboptimal alternatives.

2.5.1 Cumulative regret

Figure 12 presents the cumulative regret obtained by Thompson Sampling in both experimental scenarios.

For Scenario 1, the cumulative regret increases rapidly during approximately the first 175 episodes, reflecting the exploration of suboptimal routes. After this point, the curve becomes essentially flat, indicating that the algorithm has gathered sufficient evidence in favor of the optimal route. Consequently, suboptimal routes are selected only rarely, and almost no additional regret is accumulated during the remainder of the experiment.

In contrast, Scenario 2 exhibits a different behavior. Since the travel-time distributions of the two routes are very similar, the algorithm requires more observations to distinguish the optimal route, and consequently explores both routes for a longer period of time. Although the cumulative regret grows more slowly after approximately episode 200, it continues to increase throughout the experiment. This indicates that the algorithm has gathered considerable evidence in favor of the optimal route, but still occasionally explores the competing route, reflecting the remaining uncertainty.

These results are consistent with the findings presented in the previous sections and illustrates that learning becomes more challenging as the routes become more similar.

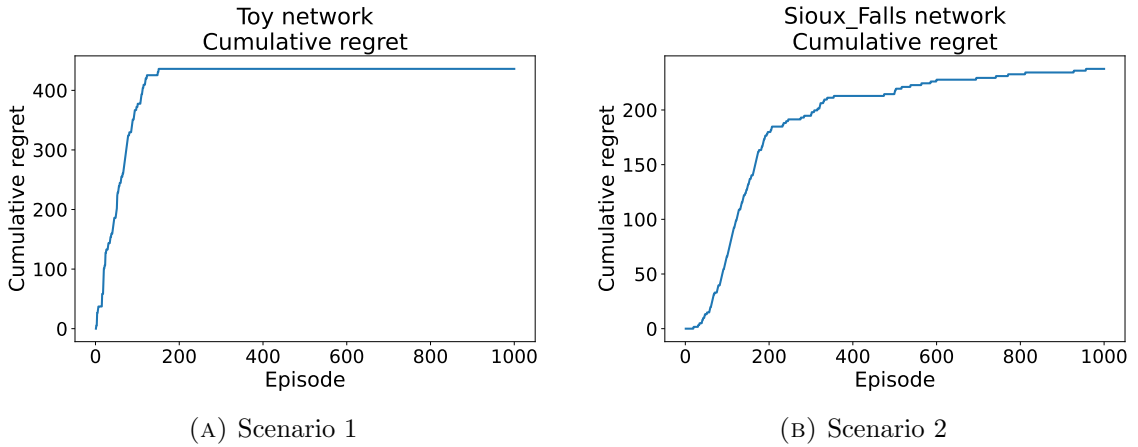


FIG. 12. Cumulative regret

2.5.2 Average regret per episode

To complement the cumulative regret analysis, it is useful to consider the average regret per episode, defined as

$$\bar{R}(t) = \frac{R(t)}{t},$$

where $R(t)$ denotes the cumulative regret up to episode t .

As the algorithm gathers evidence and increasingly selects the optimal route, the average regret per episode is expected to approach zero.

Figure 13 presents the average regret per episode for both scenarios.

For Scenario 1, the average regret per episode begins to decrease noticeably after approximately episode 150. This behavior is consistent with the cumulative regret analysis: once the algorithm has gathered sufficient evidence in favor of the optimal route, the cumulative regret grows very slowly, causing the average regret per episode to steadily decline. By the end of the experiment, the average regret reaches values close to zero, indicating that the algorithm has effectively stopped exploring suboptimal routes.

For Scenario 2, the average regret per episode starts decreasing after approximately episode 200, when the algorithm starts accumulating strong evidence in favor of the optimal route. However, unlike Scenario 1, the average regret does not approach zero as quickly, reflecting the fact that the algorithm continues to occasionally explore the suboptimal route.

An important consideration when interpreting these results is the difference in the scale of the regret values across the two scenarios. The regret values in Scenario 2 are substantially smaller than those in Scenario 1. This is because the expected travel times of the two routes are very similar, and therefore the regret incurred by selecting the suboptimal route is relatively small. Consequently, the regret magnitudes should not be directly compared across scenarios.

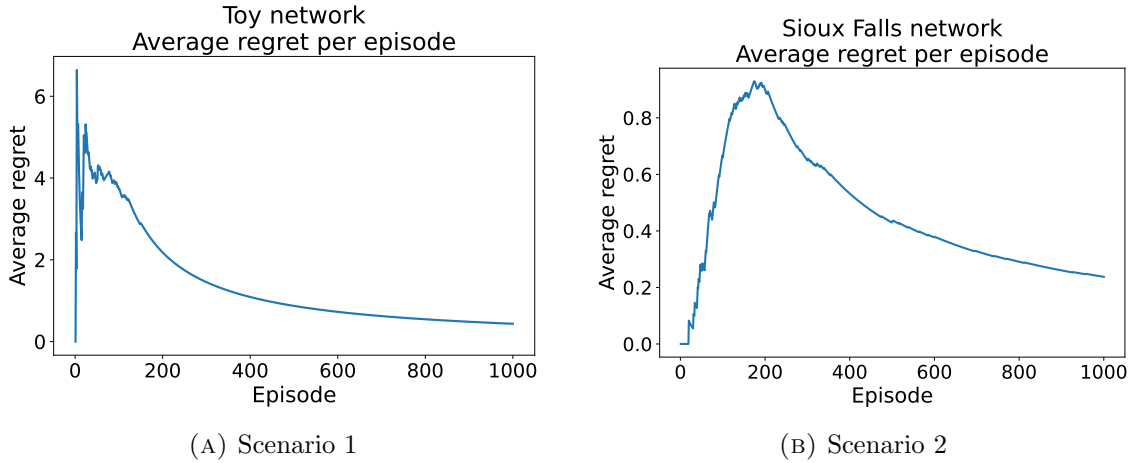


FIG. 13. Average regret per episode

3 Conclusions

This section summarizes the main conclusions of the project in a manner accessible to readers without a background in statistics. It also discusses the main difficulties encountered during the implementation of the methodology and outlines several directions for future work.

3.1 Main conclusions

The main idea behind Thompson Sampling is that, when repeatedly facing a decision among several alternatives and having little or no information about them, it is reasonable to try each alternative a few times in order to learn about its performance. Alternatives that consistently perform poorly can then be given less attention, while more promising alternatives are tried more often. Over time, more emphasis is placed on the alternatives that have shown the best performance.

This work, shows that good decisions can be made even when little information is available initially. By continuously learning from past observations Thompson Sampling is able to improve its decision over time while avoiding unnecessary exploration of clearly inferior alternatives.

3.2 Difficulties encountered

One of the main challenges encountered during this work was the computation of regret. Although Thompson Sampling does not require knowledge of the true reward distributions, the regret metric depends on the expected travel time of the optimal route. Since the expected travel times associated with each route were not available in closed form, they had to be approximated using Monte Carlo simulation.

Another challenge was related to the online nature of the learning process. Thompson Sampling updates its posterior distributions sequentially as new observations become available, rather than relying on a fixed dataset. Consequently, computational efficiency becomes an important consideration. For this reason, conjugate Bayesian models were adopted, allowing posterior distributions to be updated analytically and avoiding the need for more computationally intensive inference methods such as Markov Chain Monte Carlo (MCMC) as implemented in software such as Stan.

3.3 Future work

Several extensions of the present work could be considered in future research.

A first set of extensions concerns the statistical model used to represent travel times. One possibility would be to estimate both parameters of the Gamma distribution within the Bayesian framework, rather than fixing the shape parameter α after obtaining an initial estimate using the Method of Moments. However, this would require more computationally intensive inference methods, such as Markov Chain Monte Carlo (MCMC) implemented through software such as Stan. This approach may be difficult to reconcile with the online nature of Thompson Sampling, where posterior updates must be performed after every episode. Another possible extension would be to consider alternative travel-time models, such as the log-normal distribution. Comparing different statistical models could help assess the robustness of the proposed approach and determine whether certain models lead to improved learning behavior.

Another direction would be to consider more realistic experimental settings. This could involve applying the methodology to larger transportation networks and considering a larger set of candidate routes.

Several extensions could be made to the learning framework itself. It would be interesting to compare Thompson Sampling with other algorithms commonly used in the Multi-Armed Bandit literature, such as ϵ -greedy and Upper Confidence Bound (UCB) methods, in order to obtain a broader perspective on its strengths and weaknesses.

Another promising direction would be the study of multi-agent environments. In such settings, the travel-time distributions would no longer remain fixed. This would lead to a more realistic and dynamic learning problem.

Finally, contextual bandits could also be explored as an extension of the current framework. Instead of learning a single optimal route, the algorithm could learn the optimal route conditional on the observed context. Such contextual information could include weather

conditions, time of day, or day of the week. For example, the route with the shortest expected travel time may differ depending on whether it is raining or not.

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